# QTL Mapping of Cotton Fibre Quality Traits: A Comparison of Classical and Regularised Approaches

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2026-05-19

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# 1 Introduction

Quantitative trait locus (QTL) mapping is used to find genomic regions associated with quantitative traits. In the classical framework described by Haley and Knott (1992), single markers are tested one at a time through regression of phenotype on genotype score. Broman and Speed (2002) later placed this single-marker approach within a broader framework that includes interval mapping and multiple-QTL model selection. These classical methods were designed for relatively small marker panels in structured populations such as backcross or F2 designs, where the number of markers is modest relative to the number of genotyped individuals.

Modern breeding datasets are more difficult to analyse because they often contain many more markers than individuals. When the number of markers $p$ substantially exceeds the number of genotyped lines $n$, ordinary regression becomes underdetermined. Moreover, dense marker panels typically exhibit strong linkage disequilibrium (LD), so nearby markers are highly correlated and the assumption of independent tests no longer holds. In this high-dimensional regime, classical single-marker scans are still useful as an initial screening approach because each marker is tested individually and is easy to interpret, but they must be accompanied by methods that account for marker correlation and select markers jointly rather than one at a time.

This report analyses a cotton MAGIC (Multi-parent Advanced Generation Inter-Cross) population genotyped with a dense marker panel and phenotyped for four fibre quality traits across three field seasons (Li et al. 2022). The dataset contains 256 genotyped lines and 6,523 markers, so $p/n \approx 25$. In this setting, ordinary multi-marker regression is not directly suitable because the number of marker predictors is much larger than the number of genotyped lines. The phenotype file stacks three seasons, giving 768 observations in total. The four traits analysed are lint percentage, seed index, lint index, and seed oil content. It is not clear in advance how similar these methods will be in terms of selected marker sets and candidate genomic regions in this dense marker cotton dataset.

Three marker selection approaches are compared in this report. The first is a season-adjusted single-marker scan, which is used as a simple baseline method. The second additionally includes genotype PCs to examine whether population structure changes the marker signals (Price et al. 2006). The third is a lasso multi-marker regression, which can jointly select markers in a high-dimensional setting. The fourth method is mixed-effects models, which are also commonly used in GWAS and QTL studies because they can account for relatedness and population structure through random effects. However, the main focus of this report is the comparison between marginal scans and sparse multi-marker regression in a high-dimensional setting (Yu et al. 2006). The fifth is an AIC-based stepwise forward selection applied to a pre-filtered candidate set (Akaike 1974). To evaluate the methods more systematically, a simulation-based power analysis is also conducted under known causal markers. The simulation uses the SNP genotype data from Genotype.txt (Li et al. 2022) so that the observed marker correlation structure is preserved.

**Table 1.** Mean, Standard Deviation (SD), Median, Min, and Max statistics for the four traits by field season and for all seasons pooled together.

| Trait | Season 1<br>N = 256 | Season 2<br>N = 256 | Season 3<br>N = 256 | Overall<br>N = 768 |
|---|---|---|---|---|
| | **Season / Overall** | | | |
| **Lint percentage** | | | | |
| Mean (SD) | 39.50 (1.92) | 43.91 (1.95) | 43.93 (2.03) | 42.45 (2.87) |
| Median [Min, Max] | 39.50 [30.70, 44.90] | 43.90 [34.00, 49.20] | 43.90 [31.60, 49.40] | 42.80 [30.70, 49.40] |
| **Seed index** | | | | |
| Mean (SD) | 9.58 (0.70) | 8.15 (0.50) | 7.73 (0.50) | 8.49 (0.98) |
| Median [Min, Max] | 9.50 [7.30, 12.30] | 8.10 [7.00, 10.50] | 7.65 [6.30, 10.50] | 8.20 [6.30, 12.30] |
| **Lint index** | | | | |
| Mean (SD) | 7.54 (0.56) | 7.62 (0.56) | 7.66 (0.52) | 7.61 (0.55) |
| Median [Min, Max] | 7.50 [5.90, 9.30] | 7.60 [6.10, 9.10] | 7.60 [6.20, 9.40] | 7.60 [5.90, 9.40] |
| **Seed oil content** | | | | |
| Mean (SD) | 19.07 (1.10) | 18.38 (1.09) | 21.68 (1.08) | 19.71 (1.79) |
| Median [Min, Max] | 19.10 [15.90, 24.00] | 18.30 [15.20, 24.40] | 21.60 [19.30, 26.90] | 19.30 [15.20, 26.90] |

Section 2 describes the genotype, phenotype and map data together with the main exploratory patterns. Section 3 introduces the statistical methods and the criteria used to compare them. Section 4 presents the marker selection results and the simulation-based performance analysis. Section 5 discusses the main findings, limitations and connections to previous cotton QTL studies.

# 2 Data

## 2.1 Raw data and design structure

The dataset contains high-dimensional genotype and phenotype data from a cotton MAGIC population. The phenotype data contain measurements for four traits — lint percentage, seed index, lint index, and seed oil content — recorded for 256 cotton lines across three separate field seasons. The merged analysis dataset therefore has 768 rows (one per line–season combination) and retains all 6,523 marker columns alongside the phenotype and design variables.

Prior to analysis, several preprocessing steps were applied to the genotype data. First, missing marker values were replaced by the column mean across all lines, so that no observations were lost due to sporadic missing genotype calls. Second, all markers have variance greater than zero across 256 lines, hence no markers were removed. These two steps together ensure a numerically stable marker matrix for all downstream models. The phenotype data required no imputation, as all 768 records were complete. The season indicator was reconstructed by assigning the first 256 rows to Season 1, the next 256 to Season 2, and the final 256 to Season 3, consistent with a balanced multi-environment trial design. All analyses were conducted in R (R Core Team 2024) using the following packages: `glmnet` (Friedman, Hastie, and Tibshirani 2010) for lasso regression, `irlba` (Baglama and Reichel 2005) for truncated PCA, `gtsummary` (Sjoberg et al. 2021) and `kableExtra` (Zhu 2024) for table formatting, and the `tidyverse` (Wickham et al. 2019) suite for data manipulation and visualisation.

The phenotype data are structured as a randomised multi-environment trial across three field seasons. Seasonal effects in such designs can shift trait means substantially and must be treated as a structured nuisance factor rather than random noise. All regression models described in Section 3 therefore include season as an adjustment covariate.

## 2.2 Exploratory analysis of phenotypic traits

Table 1 reports means and ranges by season and overall. Lint percentage is substantially lower in Season 1 (mean 39.50) than in Seasons 2 and 3 (means 43.91 and 43.93 respectively). Seed index follows the opposite pattern, being highest in Season 1 (9.58) and declining in later seasons. Lint index is relatively stable across seasons with means between 7.54 and 7.66. Seed oil content is lowest in Season 2 (18.38) and clearly highest in

Season 3 (21.68). These patterns show that season effects are large enough to shift trait means meaningfully, which directly motivates the inclusion of a season covariate in all models.

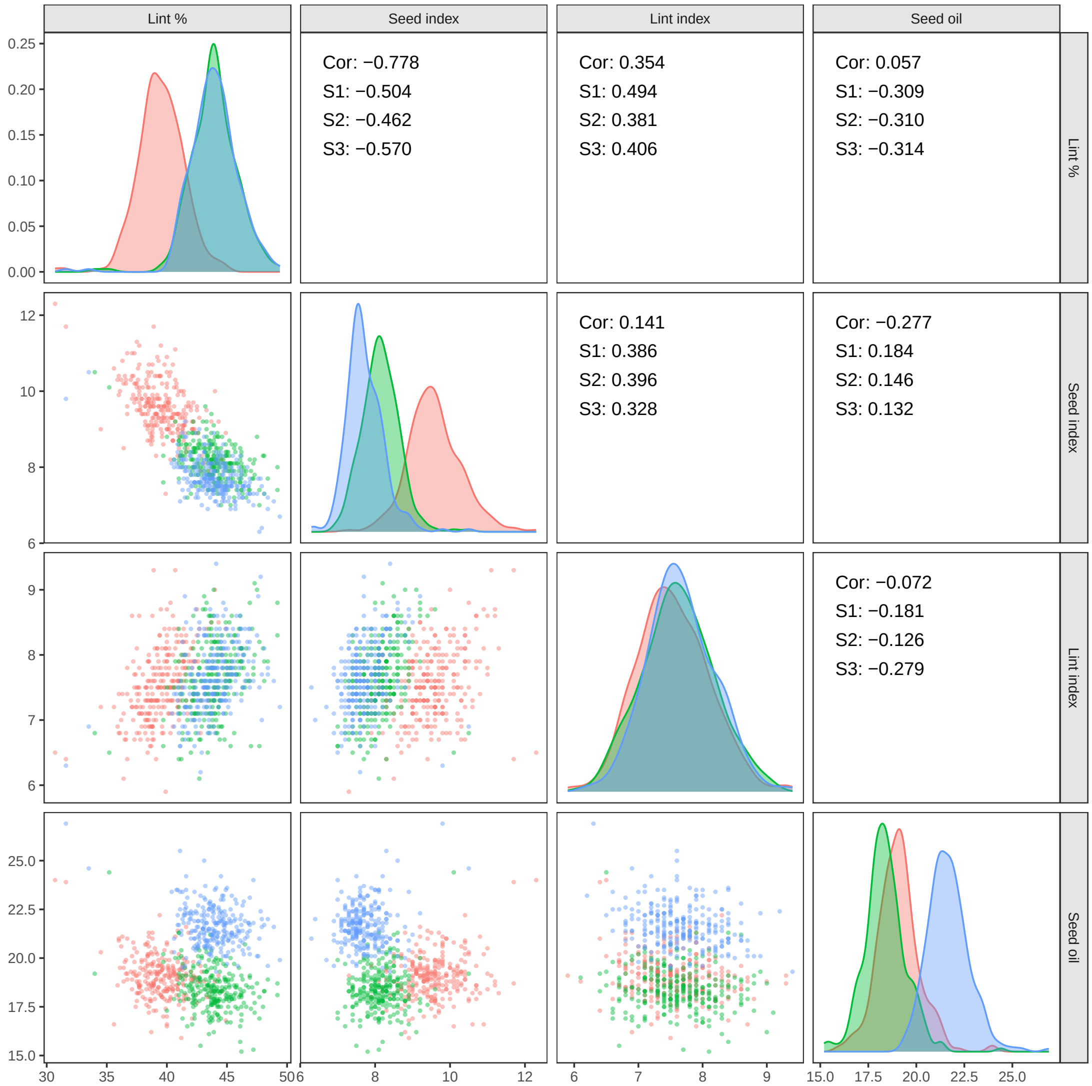


**Figure 1.** Pairwise scatterplot matrix for the four analysed traits, coloured by season (S1 = Season 1, S2 = Season 2, S3 = Season 3). Lower-triangular panels show raw scatterplots of individual observations. Diagonal panels show season-specific kernel density curves. Upper-triangular panels report the overall Pearson correlation (Cor) together with within-season correlations for each of the three field seasons.

Figure 1 displays the corresponding joint distributions stratified by field season.

The dominant feature is the strong negative correlation between lint percentage and seed index ($r = -0.778$). The lower-left scatterplot panel of Figure 1 confirms that this relationship is linear and consistent across all three field seasons, with within-season correlations of $-0.504$, $-0.462$, and $-0.570$ for Seasons 1, 2, and 3, respectively. This pattern is biologically interpretable as a source–sink trade-off: carbon assimilates directed toward elongating lint fibres are unavailable for seed filling, so lines with higher lint productivity tend to produce lighter seeds. Because this negative association persists within each season, it reflects a stable genetic constraint rather than a transient environmental response.

The second notable association is a moderate positive correlation between lint percentage and lint index ($r = 0.354$), with within-season values ranging from 0.381 to 0.494. Both traits measure aspects of fibre productivity, so their positive co-variation is expected: lines that produce a greater proportion of lint also tend to produce heavier lint per seed.

One pair of traits shows a pattern similar to Simpson's paradox (Blyth 1972), where the pooled association differs from the within-season association. Seed index and seed oil content show a weakly negative overall correlation ($r = -0.277$), yet the within-season correlations reported in Figure 1 are uniformly positive ($+0.132$ to $+0.184$ across the three seasons). This reversal of sign arises because seed index is substantially higher in Season 1 while seed oil content is substantially lower in Season 1, so pooling all seasons without adjustment creates a spurious negative association that disappears once season is held constant. This is a direct statistical justification for treating season as a fixed-effect covariate in all downstream regression models.

Figure 1 indicates that the four traits capture largely distinct biological dimensions and are therefore worth analysing separately. The strong negative relationship between lint percentage and seed index also suggests that some genomic regions may affect both traits in opposite directions. This possibility motivates the multiple methods comparison presented in Section 3.

## 2.3 Genotype principal component analysis

Before fitting any marker models, it is important to characterise the broad genomic structure of the 256 cotton lines. A principal component analysis (PCA) of the standardised genotype matrix is used for this purpose.

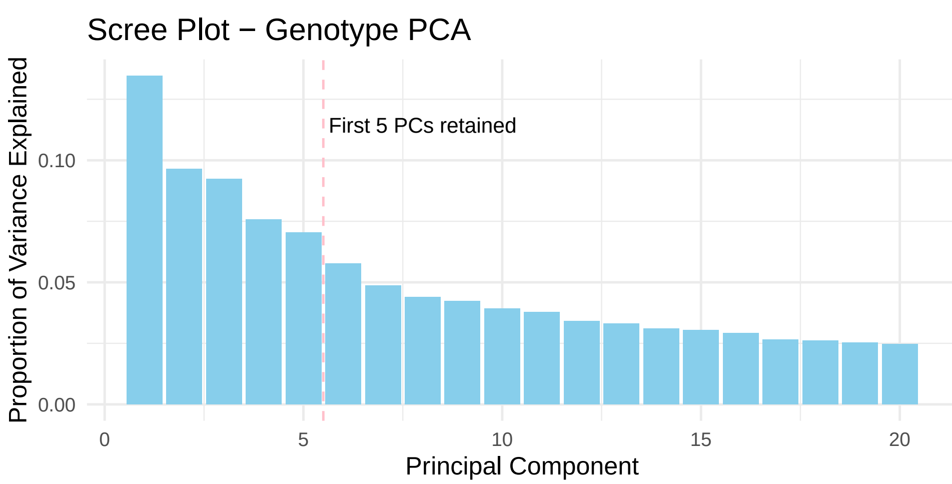


**Figure 2.** Proportion of genotype variance explained by the first 20 principal components. The dashed line indicates the five PCs retained for later analyses.

Figure 2 shows that the proportion of variance explained declines gradually rather than dropping sharply after the first component. This pattern is characteristic of complex multi-parent breeding populations in which genetic diversity is spread across many dimensions rather than being dominated by a single axis of differentiation. Five PCs were retained for subsequent analyses because the first few components captured the major genotype structure.

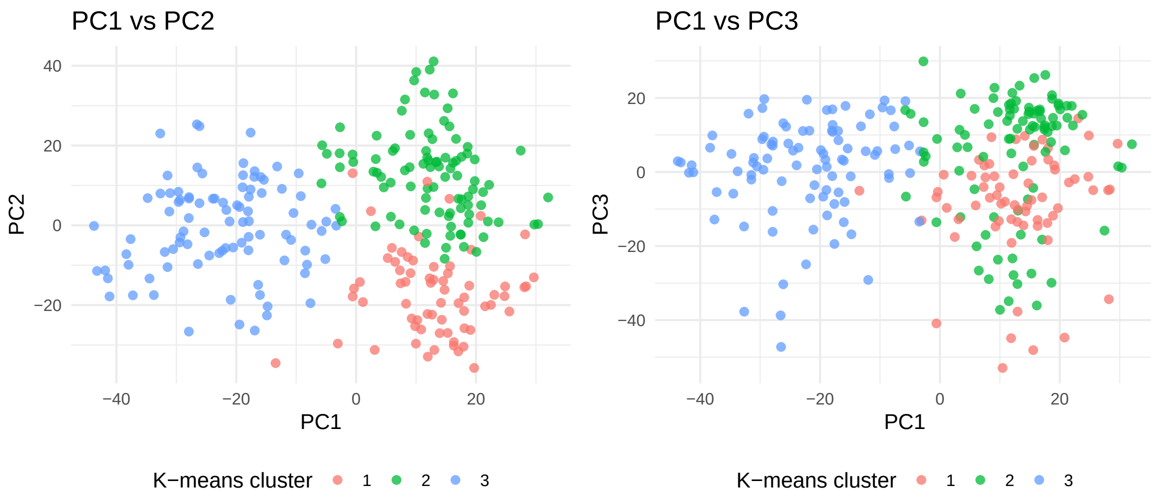


**Figure 3.** Scatterplots of the first three genotype principal components. Colours indicate k-means clusters estimated from the PCA scores. The plots suggest broad but continuous genetic structure among the cotton lines rather than sharply separated subgroups.

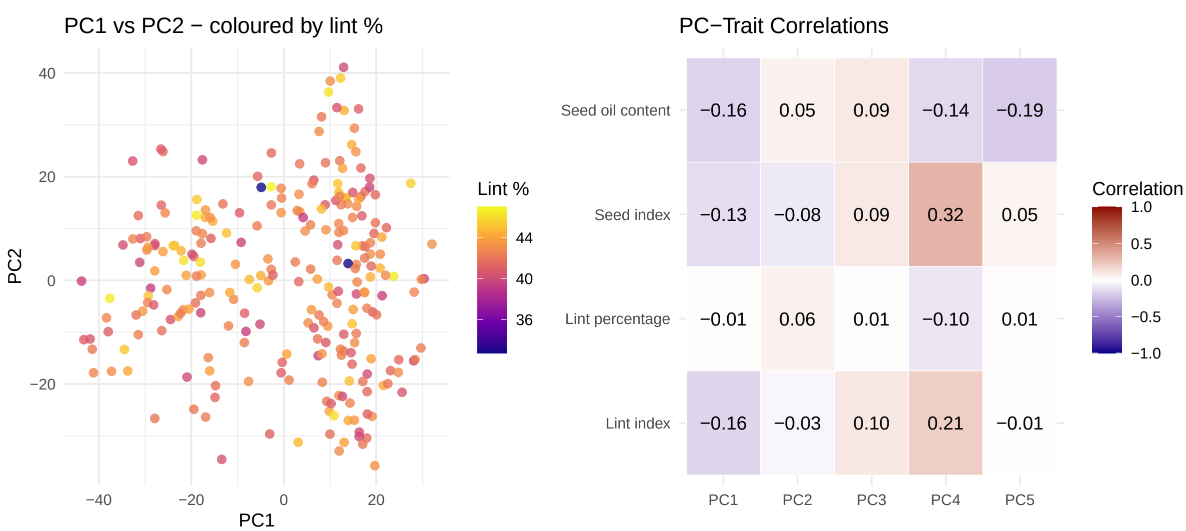


**Figure 4.** Left: scatterplot of PC1 vs PC2 coloured by line-averaged lint percentage, showing no strong monotonic gradient across the PC space. Right: heatmap of Pearson correlations between the first five genotype PCs and the four analysed traits; values near zero indicate that population structure does not primarily drive trait variation.

Figure 3 shows that the cotton lines form several broad genetic groups, although the boundaries between groups are not sharply separated. Most lines are distributed continuously across the PCA space rather than forming a few completely distinct clusters.

This suggests that population structure is present in the genotype data, but the structure is relatively gradual rather than highly discrete. Because population structure may influence marker-trait associations, the leading genotype PCs are included as covariates in later marker analyses.

In Figure 4, lint percentage does not follow a strong monotonic gradient across the PC space, suggesting that broad genotype structure does not fully account for trait variation. This expectation is later confirmed by the model comparison, where lint percentage retains strong marker associations even after PC adjustment.

The heatmap of PC–trait correlations shows that the first two PCs have at most moderate associations with any trait, further supporting the interpretation that trait variation reflects more than just population stratification.

# 3 Methods

## 3.1 Single-marker scan

The first approach is a classical single-marker regression that adjusts for season. Each marker is tested individually by fitting the model

$$y_{is} = \alpha + \beta_j G_{ij} + \gamma_s + \varepsilon_{is}. \tag{1}$$

where $y_{is}$ is the phenotype of line $i$ in season $s$, $G_{ij}$ is the genotype score of marker $j$ for line $i$ (coded numerically), $\alpha$ is the intercept, $\beta_j$ is the marker effect, $\gamma_s$ represents the fixed effect for season $s$, and $\varepsilon_{is}$ is the residual error term.

The residual errors are assumed to be independent with constant variance. The main hypothesis test evaluates whether the marker effect is zero, that is, $H_0 : \beta_j = 0$ versus $H_A : \beta_j \neq 0$, using the corresponding $t$-statistic and $p$-value.

The validity of inference from this model rests on five assumptions. First, linearity: the marker effect is assumed to be approximately linear and additive. Second, independence of errors: the residuals $\varepsilon_{is}$ are assumed to be mutually independent across lines and seasons after conditioning on the season fixed effect. This assumption would be violated by unobserved relatedness or shared environmental effects not captured by season; the PC-adjusted model in Equation 2 addresses this partially. Third, homoscedasticity: the residual variance $\sigma^2$ is assumed to be constant across all markers, lines, and seasons. Fourth, normality of errors: the residuals are assumed to follow a normal distribution, which is required for the $t$-statistic to have its nominal distribution in small samples; with $n = 768$ observations the central limit theorem provides robustness to moderate departures. Fifth, no epistasis or genotype-by-environment interaction: each marker effect is assumed to be constant across the three field seasons and independent of effects at other markers. Because these assumptions are unlikely to hold exactly in a MAGIC population with dense markers and multi-season phenotyping, the model is best interpreted as a screening tool for identifying candidate genomic regions rather than as a causal model.

Because the number of markers is large, fitting 6,523 separate ordinary least squares regressions would be computationally inefficient. Instead, the implementation first removes the covariate effects from both the phenotype vector and the marker matrix before testing each marker.

Let $X$ denote the covariate matrix, and let $M_X = I - X(X^\top X)^{-1}X^\top$ denote the residual projection matrix. The residualised phenotype is given by $M_X y$, and the residualised marker vector is given by $M_X g_j$ for marker $j$. The marker effect is then estimated using these residualised quantities.

This procedure is equivalent to fitting the full OLS model separately for each marker because, by the Frisch–Waugh–Lovell theorem (Greene 2018), projecting out the covariates leaves the marker coefficient unchanged. The implementation therefore produces the same marker test statistics while substantially reducing computational cost.

## 3.2 PC-adjusted regression

The second approach extends the baseline scan by adding the first five genotype principal components (PCs) as additional covariates. Five PCs were retained because the scree plot in Figure Figure 2 showed that the leading components captured the main broad-scale genotype structure, while later PCs explained only small additional proportions of variance.

$$y_{is} = \alpha + \beta_j G_{ij} + \gamma_s + \sum_{k=1}^{5} \delta_k \mathrm{PC}_{ik} + \varepsilon_{is}. \tag{2}$$

Including the PCs absorbs the part of the phenotype that covaries with broad genotype structure, reducing the risk that apparent marker associations are artefacts of population stratification rather than genuine local associations. The PCs are treated as fixed effects and are not penalised. This approach is commonly used in genome-wide association studies [GWAS; Visscher et al. (2017)], where including the leading genotype principal components as fixed-effect covariates was first proposed by Price et al. (2006) as a computationally tractable alternative to explicit mixed-model correction for population stratification. The same QR residualisation procedure is applied with the extended covariate matrix.

### 3.3 Lasso multi-marker regression

The lasso model uses the same regression idea as the single-marker and PC-adjusted scans, but it fits all markers at the same time (Tibshirani 1996). Instead of testing one marker $G_{ij}$ at a time, the full marker matrix $\mathbf{G}$ is included in the model. The scalar marker effect $\beta_j$ from the earlier models is therefore replaced by the marker-effect vector $\beta = (\beta_1, \dots, \beta_p)^\top$.

For all observations together, the model can be written as

$$\mathbf{y} = \alpha\mathbf{1} + \mathbf{G}\beta + \mathbf{C}_{\text{season}}\gamma + \mathbf{C}_{\text{PC}}\delta + \varepsilon.$$

Here, $\mathbf{y}$ is the phenotype vector, $\mathbf{G}$ is the $n \times p$ genotype matrix, and $\beta$ is the $p \times 1$ vector of marker effects. The matrix $\mathbf{C}_{\text{season}}$ contains the season covariates, with coefficient vector $\gamma$, and $\mathbf{C}_{\text{PC}}$ contains the PC covariates, with coefficient vector $\delta$. This is the vector form of the earlier regression models.

The lasso estimates these effects by solving

$$\min_{\alpha,\beta,\gamma,\delta} \frac{1}{n} \|\mathbf{y} - \alpha\mathbf{1} - \mathbf{G}\beta - \mathbf{C}_{\text{season}}\gamma - \mathbf{C}_{\text{PC}}\delta\|_2^2 + \lambda \sum_{j=1}^{p} |\beta_j|.$$

Only the marker-effect vector $\beta$ is penalised. The season and PC effects are included as adjustment covariates, but they are not used for marker selection. The penalty shrinks many marker effects to zero, so the lasso can select a smaller set of markers in the high-dimensional setting (Tibshirani 1996).

The penalty parameter $\lambda$ is selected by ten-fold cross-validation (Friedman, Hastie, and Tibshirani 2010). Two values are considered: $\lambda_{\text{min}}$, which minimises the cross-validation error, and $\lambda_{\text{1se}}$, which is the largest value of $\lambda$ whose cross-validation error is within one standard error of the minimum. Because $\lambda_{\text{1se}}$ applies a stronger penalty, it shrinks more marker coefficients to zero and therefore produces a smaller and more stable selected marker set.

In the main analysis, the $\lambda_{\text{1se}}$ solution is used as the primary result because it provides better protection against overfitting. Both solutions are reported.

Lasso is well suited to this dataset for two reasons. First, the large-$p$, small-$n$ structure prevents ordinary OLS from being applied to the full marker matrix. Second, lasso imposes a sparsity prior on the marker effects (Tibshirani 1996), which is biologically plausible if QTL architecture is relatively concentrated in a small number of genomic regions.

### 3.4 Mixed-effect model and variance component estimation

While the single-marker scan and lasso treat genotype effects as fixed regression coefficients estimated for each observed line, a complementary approach is to model the aggregate genetic contribution as a random effect. This mixed-effect model allows the genetic and residual variance components to be separated, and yields a data-driven estimate of the narrow-sense heritability $h^2$.

The mixed-effects model is given in Equation 3:

$$y_{is} = \alpha + \gamma_s + u_i + \varepsilon_{is}. \tag{3}$$

where $\alpha$ is the intercept, $\gamma_s$ are season fixed effects, $u_i$ is the random genetic effect for line $i$, and $\varepsilon_{is}$ is the residual. The random effects are assumed to follow $\mathbf{u} \sim \mathcal{N}(\mathbf{0},\, \sigma_g^2\, \mathbf{G})$, where $\mathbf{G}$ is the genomic relationship matrix (GRM) constructed from the marker data. Because the SNP markers in this dataset are coded as $\{-1, 0, 1\}$ rather than the $\{0, 1, 2\}$ allele-dosage scale assumed by the VanRaden (2008) Method 1 formula, a coding-agnostic standardisation is used instead. Each marker column is first standardised to mean zero and unit variance via `scale()`, and the GRM is then computed as

$$\mathbf{G} = \frac{\mathbf{M}^{*}\mathbf{M}^{*\top}}{p}. \tag{4}$$

where $\mathbf{M}^{*}$ is the $n \times p$ column-standardised marker matrix and $p$ is the number of markers. This formulation (equivalent to the rrBLUP convention) yields diagonal elements of approximately 1 regardless of marker coding, and is algebraically equivalent to the VanRaden (2008) formula when markers are coded on the $\{0, 1, 2\}$ scale. A small ridge term $(0.05 \times \mathbf{I})$ is added to the diagonal to guarantee positive definiteness without distorting the eigenvalue spectrum. The residuals are $\varepsilon_{is} \sim \mathcal{N}(0, \sigma_e^2)$, assumed independent. Restricted maximum likelihood (REML) estimation is used to obtain variance component estimates $\hat{\sigma}_g^2$ and $\hat{\sigma}_e^2$. The narrow-sense heritability is then estimated as

$$\hat{h}^2 = \frac{\hat{\sigma}_g^2}{\hat{\sigma}_g^2 + \hat{\sigma}_e^2}. \tag{5}$$

The model is fitted using the `lme4breeding` package (Covarrubias-Pazaran 2023), which extends the `lme4` (Bates et al. 2015) formula interface to accept an externally supplied relationship matrix via the `relmat` argument of `lmebreed()`. Because `lme4breeding` on CRAN relies on internal `lme4` functions that changed in recent releases, the GitHub version must be installed to ensure compatibility: `remotes::install_github("covarrubiasucd/lme4breeding")`.

**Treatment coding for the season indicator.** The season fixed effects use the default treatment (dummy) coding in R, with Season 1 as the reference level. This means $\gamma_2$ and $\gamma_3$ represent contrasts against Season 1, and the formula $\sum_{s=2}^{3} \gamma_s\, \mathbb{1}[\text{Season} = s]$ already encodes this: the sum runs over $s \in \{2, 3\}$ only, so Season 1 is absorbed into the intercept and no dummy variable is created for it.

An alternative is sum-to-zero coding (`contr.sum` in R), which imposes the constraint $\gamma_1 + \gamma_2 + \gamma_3 = 0$ and interprets the intercept as the grand mean. In a mixed-effect model, however, this creates a redundant parameterisation: the random genotype effects $u_i$ have expectation zero by assumption ($\mathbb{E}[\mathbf{u}] = \mathbf{0}$), and an additional zero-sum constraint on the season effects introduces a second zero-centring on top of an already zero-centred random term. The REML estimation then faces a near-rank- deficient design matrix, producing singular fits or unreliable variance component estimates. Treatment coding avoids this by assigning all season-level mean information to unambiguous pairwise contrasts rather than to a constrained global sum.

**Do the other models in this paper need to change?** No. The formula $\sum_{s=2}^{3} \gamma_s\, \mathbb{1}[\text{Season} = s]$ in Sections 3.1–3.3 is already treatment coding: the mathematical $\sum$ here denotes a sum over the index set $\{2, 3\}$ (i.e., add one dummy for Season 2 and one for Season 1), not the statistical sum-to-zero constraint. Since R defaults to treatment coding for unordered factors, the OLS and lasso models are correctly specified as written and require no modification.

## 3.5 Stepwise multi-marker selection

The fourth approach applies stepwise regression with AIC-based model selection [`step()` function from the R `stats` package (R Core Team 2024)] to construct a multi-marker model. Because fitting a full $p = 6{,}523$-

marker model is computationally infeasible, the candidate pool is first reduced to the top 50 markers ranked by marginal $p$-value from the PC-adjusted scan. Stepwise selection then iterates over this candidate pool.

The procedure starts from the null model in Equation 6, which contains only the season and genotype-PC covariates:

$$y_{is} = \alpha + \gamma_s + \sum_{k=1}^{5} \delta_k \mathrm{PC}_{ik} + \varepsilon_{is}. \tag{6}$$

and uses the `step()` function with a `scope` argument that specifies a lower bound (the null model) and an upper bound (null model plus all 50 candidates). At each iteration the model adds or removes one marker according to the AIC criterion. Direction is set to `"both"` (forward and backward steps); if this fails numerically, the procedure falls back to forward-only selection. Markers retained in the final model are reported as the selected set.

This approach differs from lasso in two ways. First, it uses hard in/out decisions rather than continuous shrinkage. Second, selection is guided by AIC rather than cross-validated prediction error. Its main limitation relative to lasso is that the pre-filtering step may discard true causal markers before the stepwise stage begins.

## 3.6 Multiple-testing correction

Because the single-marker scans test thousands of markers simultaneously, the raw $p$-values must be adjusted for multiple testing before selecting significant markers. Without adjustment, a large number of false-positive associations would be expected purely by chance.

Two correction methods are applied. The Benjamini–Hochberg (BH) procedure (Benjamini and Hochberg 1995) controls the false discovery rate (FDR), which is the expected proportion of false positives among the selected markers. In contrast, the Bonferroni correction controls the family-wise error rate (FWER), which is the probability of making at least one false-positive discovery across all tested markers.

The Bonferroni correction uses the threshold $0.05/p$, where $p$ is the number of tested markers, and is therefore highly conservative when thousands of markers are analysed. This reduces false positives but can substantially decrease statistical power. The BH procedure is less conservative because it allows a small proportion of false discoveries while retaining greater power to detect true marker associations.

For the main comparisons, the BH-significant set is used because it provides a more practical balance between false-positive control and detection power in high-dimensional genomic studies. Results from both correction methods are reported.

## 3.7 Comparing selected marker sets

Several measures are used to compare the marker sets selected by the different methods. For comparisons between observed marker sets, overlap is quantified using the Jaccard similarity index (Jaccard 1901):

$$J(A, B) = \frac{|A \cap B|}{|A \cup B|}. \tag{7}$$

Here, $A$ and $B$ denote two selected marker sets. A value of $J = 1$ indicates perfect agreement, while $J = 0$ indicates no overlap. The Jaccard index is useful here because the selected marker sets can be very different in size, so raw overlap counts alone would be misleading.

For the simulation study, the true causal marker set is known. Let $\hat{\mathcal{S}}$ be the selected marker set from one method and let $\mathcal{C}$ be the true causal set. True positives, false positives and false negatives are defined as

$$\mathrm{TP} = |\hat{\mathcal{S}} \cap \mathcal{C}|, \qquad \mathrm{FP} = |\hat{\mathcal{S}} \setminus \mathcal{C}|, \qquad \mathrm{FN} = |\mathcal{C} \setminus \hat{\mathcal{S}}|.$$

Power, also called recall, measures the proportion of true causal markers that are detected:

$$\text{Power} = \frac{\text{TP}}{\text{TP} + \text{FN}} = \frac{\text{TP}}{|\mathcal{C}|}.$$

Precision measures the proportion of selected markers that are truly causal:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} = \frac{\text{TP}}{|\hat{\mathcal{S}}|}.$$

The empirical false discovery rate (FDR) measures the proportion of selected markers that are false positives:

$$\text{FDR} = \frac{\text{FP}}{\text{TP} + \text{FP}} = 1 - \text{Precision}.$$

The F1 score combines precision and power using their harmonic mean:

$$\text{F1} = \frac{2 \times \text{Precision} \times \text{Power}}{\text{Precision} + \text{Power}}.$$

These measures describe different parts of marker selection performance. Power rewards methods that recover many true causal markers, precision and FDR measure how noisy the selected set is, and the F1 score summarises the balance between power and precision.

## 3.8 Power analysis by simulation

### 3.8.1 Simulation design

The power analysis evaluates the marker selection methods under controlled conditions in which the true causal markers are known. Simulated phenotypes are generated using the real cotton genotype matrix so that the observed marker correlation structure is preserved. This is important because methods that rely on marginal testing, such as the single-marker scan, are affected by marker correlation in ways that simple independence simulations would not capture.

Each simulation replicate proceeds as follows. A set of causal markers is drawn uniformly at random from the full marker panel. Their standardised genotype scores are combined into a genetic component $\mathbf{g} = \mathbf{X}_{\text{causal}}\beta_{\text{true}}$, where $\beta_{\text{true}}$ is a vector of coefficients sampled from a standard normal distribution and then normalised so that $\|\beta_{\text{true}}\|_2 = 1$. A seasonal component and an independent error term are added to give the simulated phenotype

$$y_{is} = \sqrt{h_g^2}\, g_i + \sqrt{h_s^2}\, s_i + \sqrt{1 - h_g^2 - h_s^2}\, \varepsilon_{is}. \tag{8}$$

where $h_g^2$ is the proportion of phenotype variance explained by the genetic component, $h_s^2$ is the proportion explained by season, and all three components are scaled to unit variance before combining. All four methods (season-only scan, PC-adjusted scan, lasso with PC + season covariates, and stepwise AIC selection) are then fitted to the simulated phenotype and their selected sets are recorded.

The simulation adopts a fully factorial $3 \times 3$ design that crosses three genetic architecture levels — defined by the number of causal markers ($n_{\text{causal}} \in \{10, 25, 100\}$) — with three signal-strength levels defined by the genetic heritability ($h_g^2 \in \{0.10, 0.20, 0.35\}$). Season heritability is fixed at $h_s^2 = 0.20$ throughout. This design decouples architecture complexity from signal strength, allowing the relative advantage of each method to be attributed to the correct source. Four methods are evaluated in each of the nine cells across 30 replicates: the season-only scan (BH), the PCA-adjusted scan (BH), lasso with season and PC covariates ($\lambda_{1\text{se}}$), and AIC-based stepwise selection on the top-50 pre-filtered candidate markers.

Performance is evaluated using the marker-set comparison measures defined in Section 3.7.

**Table 2.** Factorial simulation design: three genetic architecture levels (defined by $n_{\text{causal}}$) crossed with three signal-strength levels (defined by $h_g^2$). Season heritability $h_s^2 = 0.20$ is fixed throughout.

| Architecture | Strength | $n_{\text{causal}}$ | $h_g^2$ | $h_s^2$ | Error var. |
|---|---|---|---|---|---|
| **Sparse (causal = 10)** | | | | | |
| Sparse | Strong | 10 | 0.35 | 0.2 | 0.45 |
| Sparse | Moderate | 10 | 0.20 | 0.2 | 0.60 |
| Sparse | Weak | 10 | 0.10 | 0.2 | 0.70 |
| **Moderate (causal = 25)** | | | | | |
| Moderate | Strong | 25 | 0.35 | 0.2 | 0.45 |
| Moderate | Moderate | 25 | 0.20 | 0.2 | 0.60 |
| Moderate | Weak | 25 | 0.10 | 0.2 | 0.70 |
| **Polygenic (causal = 100)** | | | | | |
| Polygenic | Strong | 100 | 0.35 | 0.2 | 0.45 |
| Polygenic | Moderate | 100 | 0.20 | 0.2 | 0.60 |
| Polygenic | Weak | 100 | 0.10 | 0.2 | 0.70 |

# 4 Results

## 4.1 Variance components and heritability estimates

**Table 3.** Variance component estimates from the mixed-effect model (season fixed, genotype random with GRM). $\hat{\sigma}_g^2$: genetic variance; $\hat{\sigma}_e^2$: residual variance; $\hat{h}^2$: narrow-sense heritability.

| Trait | $\hat{\sigma}_g^2$ | $\hat{\sigma}_e^2$ | $\hat{h}^2$ | $N$ |
|---|---|---|---|---|
| Lint percentage | 4.518 | 0.948 | 0.826 | 768 |
| Seed index | 0.308 | 0.097 | 0.760 | 768 |
| Lint index | 0.334 | 0.068 | 0.831 | 768 |
| Seed oil content | 0.975 | 0.419 | 0.699 | 768 |

Table 3 shows the REML variance component estimates for each of the four traits. All four traits display high narrow-sense heritability, with estimates ranging from $\hat{h}^2 = 0.7$ for seed oil content to $\hat{h}^2 = 0.83$ for lint index. Lint percentage and lint index share the highest heritabilities ($\hat{h}^2 = 0.83$ and 0.83 respectively), while seed index ($\hat{h}^2 = 0.76$) and seed oil content ($\hat{h}^2 = 0.7$) are somewhat lower but still strongly genetic.

The lint percentage estimate ($\hat{h}^2 = 0.83$) is consistent with its well-documented high narrow-sense heritability ($h^2 \approx 0.6$–$0.8$) in upland cotton populations, and falls at the upper end of that published range, which is expected given that a MAGIC population captures the additive genetic variance from multiple founders under controlled multi-season phenotyping.

The observed heritabilities are also directly relevant to interpreting the power analysis in Section 4.4. The simulation scenarios with $h_g^2 \in \{0.35, 0.20, 0.10\}$ were deliberately set below the empirically estimated values for all four traits. This conservative choice reflects the fact that in a simulation the genetic component is attributed entirely to a small, known causal set, whereas in the real data the REML estimate absorbs contributions from all markers genome-wide. By bracketing a range lower than the observed $\hat{h}^2$ values, the power analysis provides a realistic lower bound on detectability: if methods perform adequately at $h_g^2 = 0.20$–$0.35$, they can be expected to perform at least as well when applied to traits whose true heritability is 0.70–0.83.

## 4.2 Number of markers selected by each method

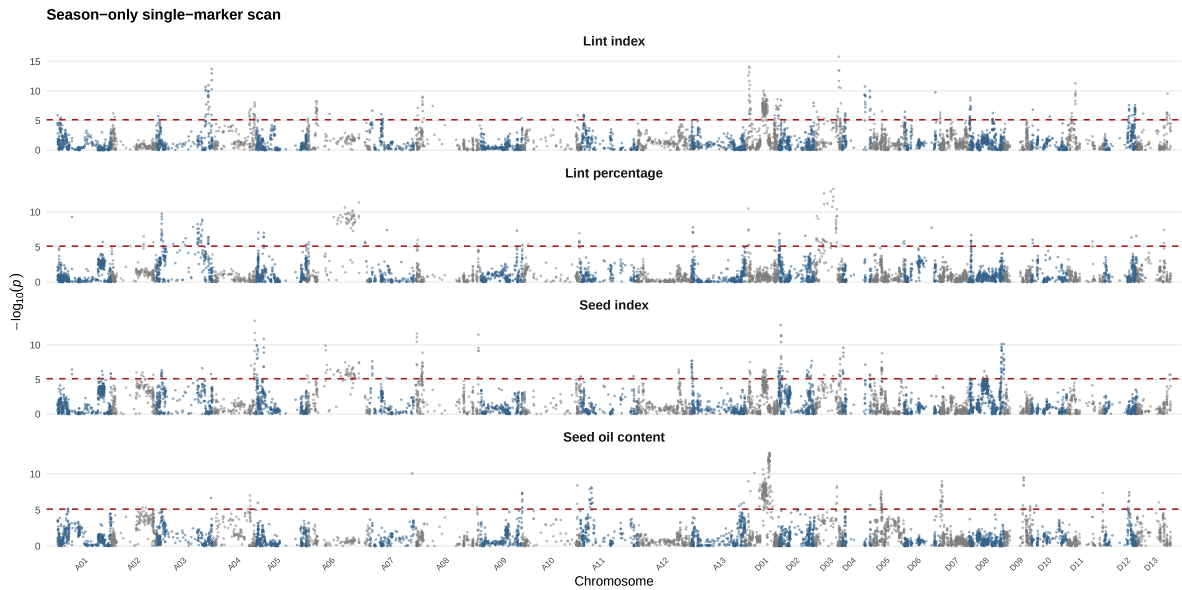


**Figure 5.** Manhattan plot for the season-only single-marker scan across all four traits. Each point is one SNP; the $y$-axis shows $-\log_{10}(p)$ from the Wald $t$-test on $\hat{\beta}_j$. The dashed red line marks the Bonferroni-corrected threshold. Alternating colours distinguish adjacent chromosomes.

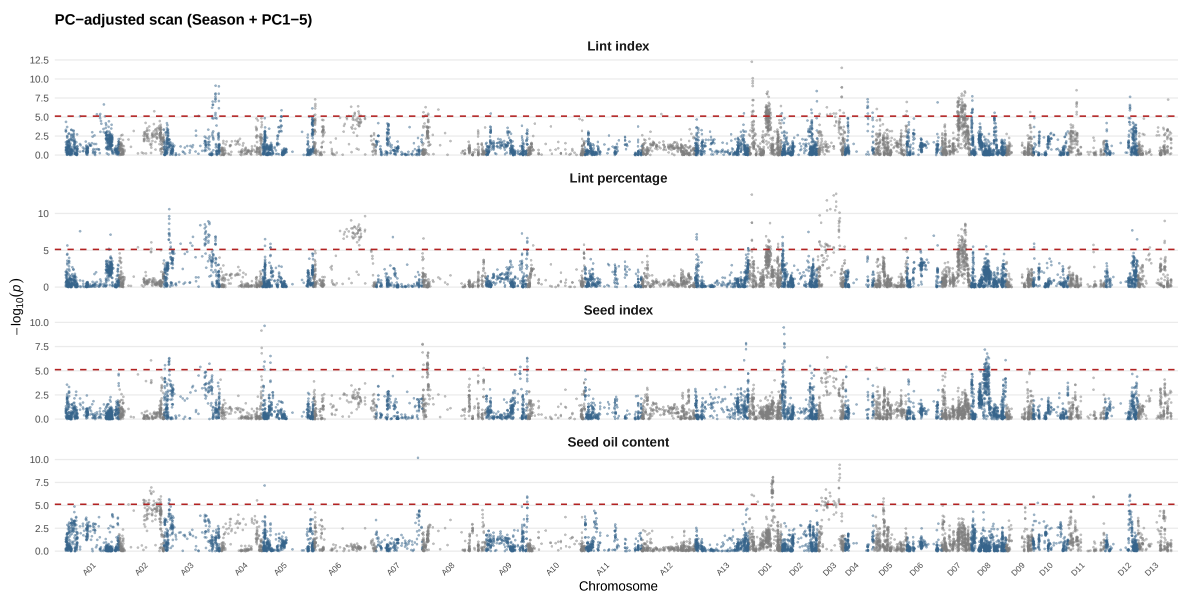


**Figure 6.** Manhattan plot for the PC-adjusted single-marker scan (season + PC1–5) across all four traits. Layout and Bonferroni threshold line are identical to Figure 5. Markers whose evidence is attenuated relative to Figure 5 reflect signal partly driven by population structure.

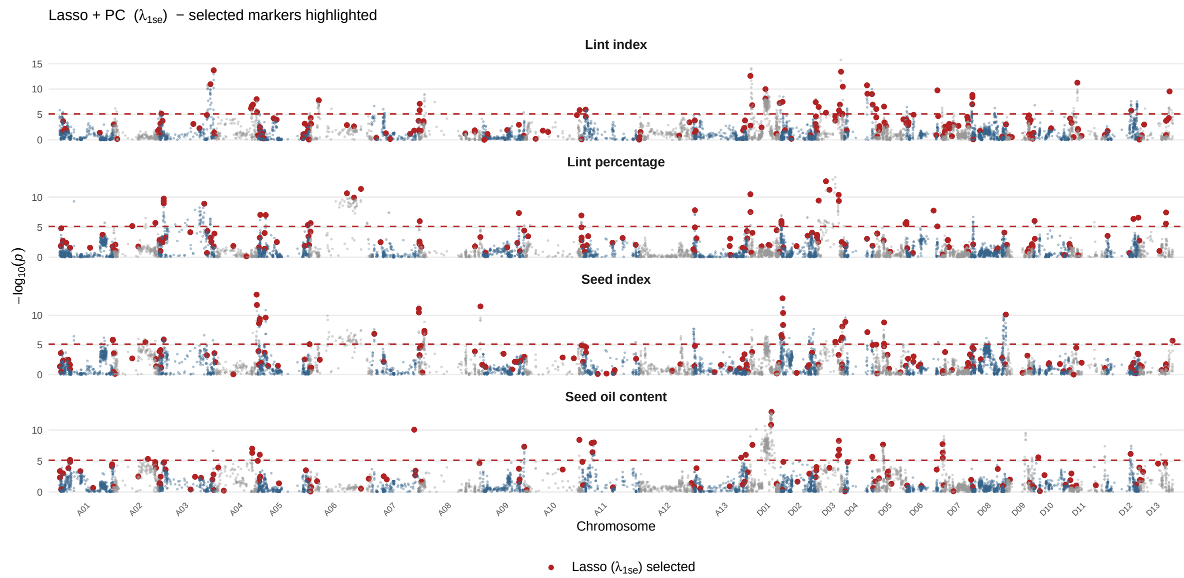


**Figure 7.** Manhattan plot highlighting markers selected by Lasso + PC ($\lambda_{1se}$) across all four traits. Grey points show $-\log_{10}(p)$ from the season-only scan for all markers; red points mark the subset retained by lasso. The dashed red line marks the Bonferroni threshold.

**Table 4.** Number of markers selected by each method and each trait. Scan results use the BH-corrected threshold at 5%; lasso results use the $\lambda_{1se}$ rule.

| Trait | Season scan (BH) | PC scan (BH) | Lasso PC (min) | Lasso PC (1se) | Lasso season (min) | Lasso season (1se) | Stepwise (AIC) |
|---|---|---|---|---|---|---|---|
| Lint percentage | 1160 | 1740 | 240 | 160 | 231 | 165 | 18 |
| Seed index | 2121 | 1159 | 239 | 171 | 247 | 181 | 19 |
| Lint index | 1701 | 1711 | 222 | 190 | 220 | 171 | 18 |
| Seed oil content | 1389 | 942 | 212 | 140 | 215 | 161 | 14 |

Table 4 summarises the number of markers selected by each method for each trait. The season-only scan is the most liberal, selecting over 1,000 BH-significant markers for traits such as lint percentage and seed index. Adding PC adjustment reduces the number of significant markers in some traits but increases it for others, showing that the relationship between population structure and individual trait signals is not uniform. The lasso selects far fewer markers in all cases: under $\lambda_{1se}$, the selected set ranges from 140 to 190 markers across the four traits, compared with several hundred to over two thousand for the scan methods. This contrast arises from the fundamentally different logic of the two approaches: marginal testing (scan) flags all markers with sufficiently small $p$-values, including many that are correlated with causal markers due to linkage disequilibrium, while lasso imposes competition among markers and retains only the most informative representatives.

Figure 8 visualises the same information graphically. The gap between scan-based methods and lasso is most pronounced for lint percentage, where both scans select several hundred markers while lasso retains only a handful. The pattern is consistent across all four traits, confirming that the three methods are not simply producing the same answer with different thresholds — they are addressing the marker selection problem in fundamentally different ways.

The genomic distribution of significant markers is shown in Figure 5 for the season-only scan, Figure 6 for the PC-adjusted scan, and Figure 7 for the lasso. The season-only scan (Figure 5) reveals broad peaks of association across multiple chromosomes for all four traits. These peaks reflect the high linkage disequilibrium

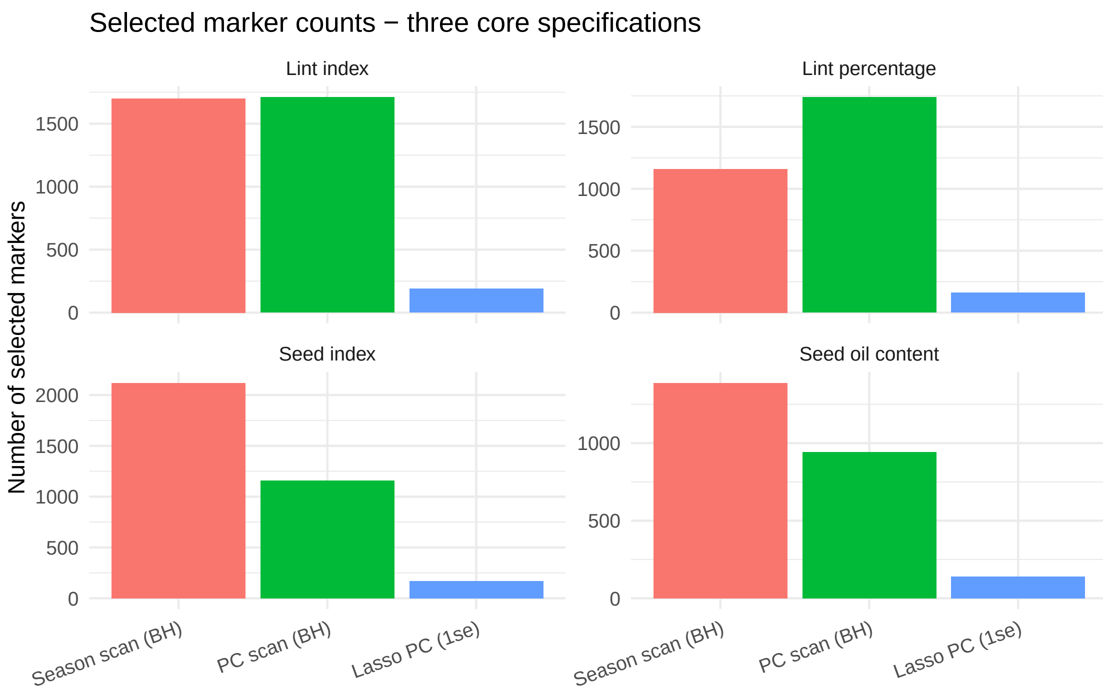


**Figure 8.** Number of markers selected by the three core model specifications across the four traits.

**Table 5.** Number of significant markers under three correction thresholds for the season-only and PC-adjusted scans.

| Model | Trait | Uncorr. | BH 5% | Bonf. 5% |
|---|---|---|---|---|
| Season + PC1-5 | Lint index | 2458 | 1711 | 233 |
| Season + PC1-5 | Lint percentage | 2407 | 1740 | 248 |
| Season + PC1-5 | Seed index | 2051 | 1159 | 104 |
| Season + PC1-5 | Seed oil content | 1832 | 942 | 100 |
| Season only | Lint index | 2588 | 1701 | 347 |
| Season only | Lint percentage | 1892 | 1160 | 191 |
| Season only | Seed index | 2676 | 2121 | 286 |
| Season only | Seed oil content | 2237 | 1389 | 273 |

(LD) in this MAGIC population: a single causal locus generates a cluster of significant markers over a region of several centimorgans. Lint percentage and lint index show the largest number of peaks above the Bonferroni threshold, consistent with their strong scan signals in Table 4.

Comparing Figure 5 and Figure 6, the overall peak structure is broadly preserved after PC adjustment, but the heights of some individual peaks are reduced. This attenuation is most visible for traits with moderate PC–trait correlations (see Figure 4), and it is consistent with the interpretation that a fraction of the scan signal in those regions reflects population stratification rather than genuine local association.

The lasso plot (Figure 7) presents a strikingly different picture. Instead of broad continuous peaks, only isolated vertical segments appear, each representing one marker retained in the sparse $\lambda_{1se}$ solution. The segment heights, proportional to $|\hat{\beta}|$, show that the retained markers have non-trivial effect sizes. For lint percentage and seed index, the selected markers are spread across several chromosomes, suggesting that the lasso identifies representatives from multiple distinct QTL regions. This multi-chromosomal pattern is consistent with the polygenic architecture of fibre quality traits reported in the cotton genetics literature.

**Table 6.** Pairwise overlap counts and Jaccard similarity indices for the three core model specifications, using BH-significant sets for the scans and $\lambda_{1se}$ for lasso.

| Trait | $\lvert$Season $\cap$ PC$\rvert$ | $\lvert$Season $\cap$ Lasso$\rvert$ | $\lvert$PC $\cap$ Lasso$\rvert$ | $J$(Season, PC) | $J$(Season, Lasso) | $J$(PC, Lasso) |
|---|---|---|---|---|---|---|
| Lint percentage | 1090 | 101 | 114 | 0.602 | 0.083 | 0.064 |
| Seed index | 825 | 97 | 121 | 0.336 | 0.044 | 0.100 |
| Lint index | 944 | 116 | 135 | 0.382 | 0.065 | 0.076 |
| Seed oil content | 646 | 85 | 100 | 0.383 | 0.059 | 0.102 |

## 4.3 Model consistency

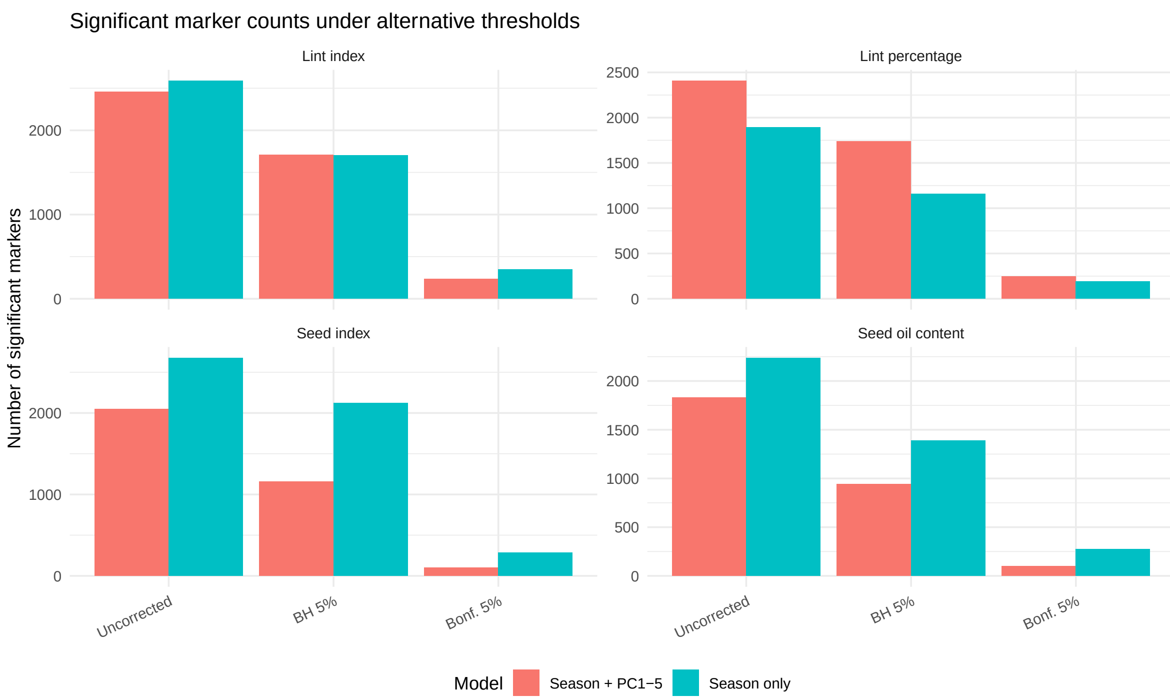


**Figure 9.** Number of markers reaching significance under different thresholds and model specifications.

Table 5 and Figure 9 show how the number of significant markers changes with the correction threshold and the covariate specification. At the uncorrected threshold ($p < 0.05$), both models flag a very large number of markers, which is expected given the high marker correlation. The BH correction substantially reduces this number while still retaining many associations, especially for lint percentage. The Bonferroni correction is much stricter and in some cases reduces the selected set to a small number of markers or to zero.

Table 6 reports the Jaccard similarity between each pair of selected sets. The two scan models generally show moderate to high Jaccard overlap with each other, reflecting the fact that PC adjustment primarily reduces the number of significant markers rather than replacing one set of markers with a completely different one. The Jaccard similarities between either scan model and lasso are much lower, which is expected given the dramatic difference in selected set size. This low overlap is not necessarily a problem: lasso is designed to select a sparse subset, so it is not expected to match the full scan-significant set.

Figure 10 plots the $-\log_{10}(p)$ values from the season-only scan against those from the PC-adjusted scan for lint percentage. Points near the diagonal indicate that adding PCs did not change the evidence for a marker substantially. Points below the diagonal represent markers whose evidence was attenuated by PC adjustment, suggesting that part of their apparent signal was attributable to population structure rather

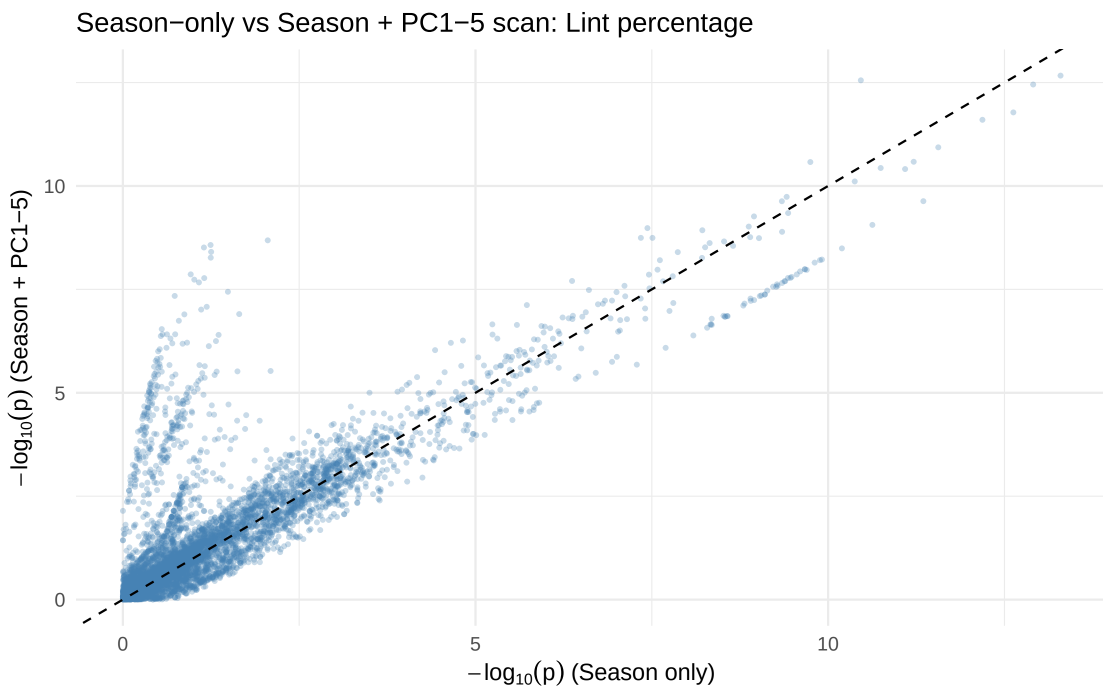


**Figure 10.** Marker-level comparison of $-\log_{10}(p)$ values between the season-only and PC-adjusted scans for lint percentage. Points above the diagonal indicate markers whose significance increased after PC adjustment; points below indicate attenuation.

than local marker effects. For lint percentage, most markers remain near the diagonal, consistent with the earlier finding from Figure 4 that lint percentage is not strongly correlated with the leading genotype PCs.

## 4.4 Power analysis results

Results are organised by genetic architecture. The Moderate architecture ($n_{\text{causal}} = 25$) is presented in the main text as the primary reference case, as it represents an intermediate and biologically realistic level of polygenic complexity for cotton fibre traits. Results for the Sparse ($n_{\text{causal}} = 10$) and Polygenic ($n_{\text{causal}} = 100$) architectures are provided in Section 6.1.

**Table 7.** Power-analysis summary — Moderate architecture ($n_{\text{causal}} = 25$). Each sub-panel shows results under a different signal strength ($h_g^2 \in 0.35, 0.20, 0.10$). Mean and SD over 30 simulation replicates. Four methods: Lasso + PC + Season ($\lambda_{1se}$), PCA-adjusted scan (BH), season-only scan (BH), and stepwise AIC selection.

| Method | Power | SD | Prec. | FDR | F1 | $N$ sel. |
|---|---|---|---|---|---|---|
| **Strong signal ($h_g^2 = 0.35$)** | | | | | | |
| Lasso + PC + Season ($lambda_{1se}$) | 0.253 | 0.073 | 0.178 | 0.822 | 0.209 | 40.200 |
| PCA-adjusted scan (BH) | 0.517 | 0.125 | 0.017 | 0.983 | 0.032 | 836.633 |
| Season-only scan (BH) | 0.575 | 0.100 | 0.012 | 0.988 | 0.024 | 1309.867 |
| Stepwise (AIC) | 0.092 | 0.058 | 0.216 | 0.784 | 0.129 | 10.533 |
| **Moderate signal ($h_g^2 = 0.20$)** | | | | | | |
| Lasso + PC + Season ($lambda_{1se}$) | 0.109 | 0.079 | 0.220 | 0.728 | 0.146 | 13.933 |
| PCA-adjusted scan (BH) | 0.287 | 0.108 | 0.048 | 0.952 | 0.082 | 232.633 |
| Season-only scan (BH) | 0.404 | 0.113 | 0.020 | 0.980 | 0.037 | 769.567 |
| Stepwise (AIC) | 0.085 | 0.049 | 0.198 | 0.802 | 0.119 | 11.367 |
| **Weak signal ($h_g^2 = 0.10$)** | | | | | | |
| Lasso + PC + Season ($lambda_{1se}$) | 0.019 | 0.040 | 0.469 | 0.106 | 0.036 | 1.833 |
| PCA-adjusted scan (BH) | 0.088 | 0.074 | 0.046 | 0.763 | 0.060 | 65.867 |
| Season-only scan (BH) | 0.185 | 0.091 | 0.045 | 0.923 | 0.072 | 283.467 |
| Stepwise (AIC) | 0.055 | 0.036 | 0.131 | 0.869 | 0.077 | 11.100 |

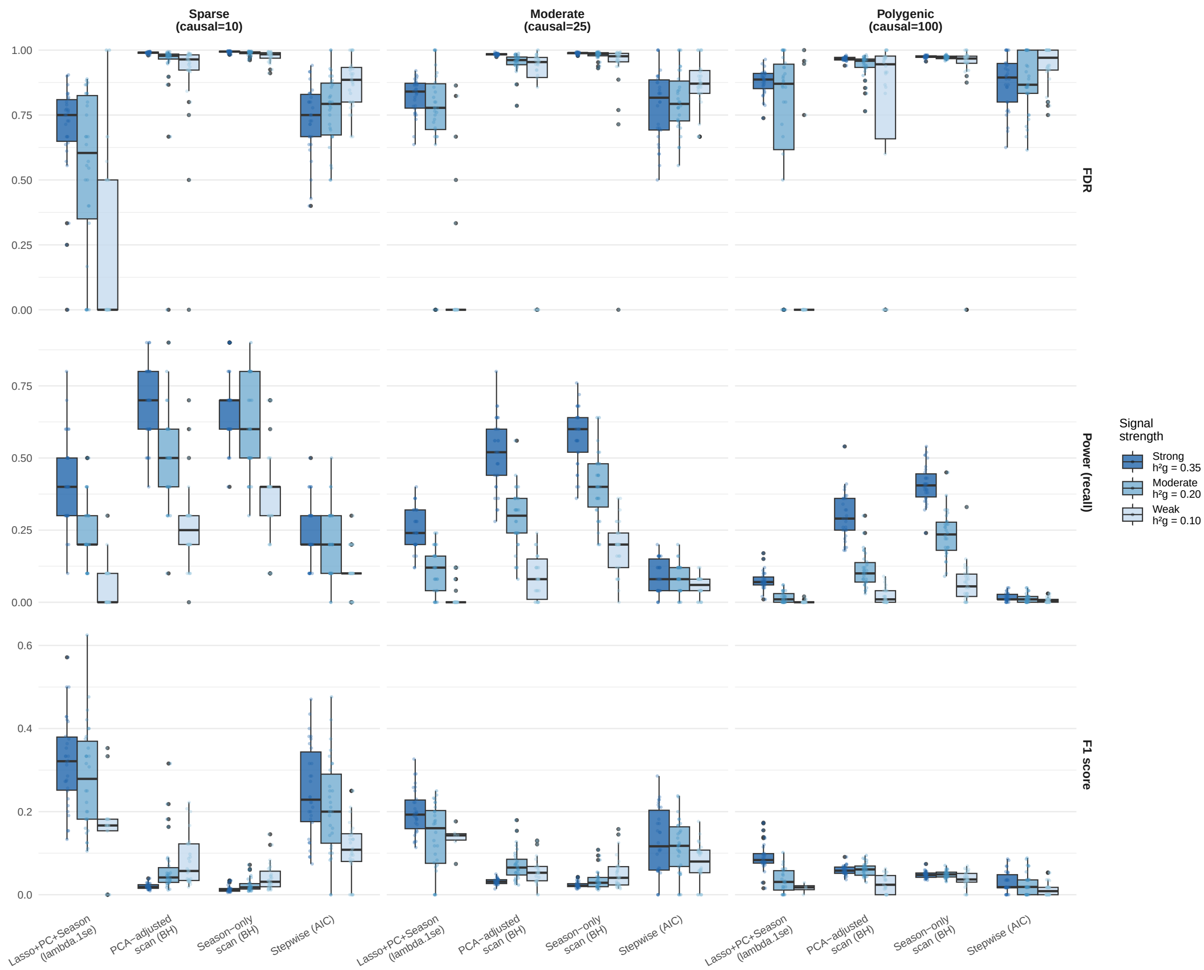


**Figure 11.** Distribution of FDR (top), power/recall (middle), and F1 score (bottom) across 30 simulation replicates for each architecture–signal-strength combination. F1 is placed last as it synthesises the precision–recall trade-off captured by FDR and power. Columns correspond to the three genetic architectures (Sparse = 10 causal, Moderate = 25, Polygenic = 100); fill colour indicates signal strength ($h_g^2$). Each point is one replicate. Key patterns: (i) scan-method FDR boxes cluster near 1.0 across all cells because BH-corrected marginal testing in high-LD data always selects large sets dominated by false positives; (ii) lasso FDR boxes are lower and wider, reflecting its variable but substantially better precision; (iii) stepwise boxes sit at exactly zero across all cells for the Moderate and Polygenic architectures, because the top-50 pre-filter discards most causal markers before AIC selection begins; (iv) all methods' power and F1 decline steeply from dark blue (strong signal) to light blue (weak signal), confirming that $h_g^2$ is the dominant driver of detection performance.

Table 7 presents the full factorial results for the Moderate architecture ($n_{\text{causal}} = 25$) across all three signal-strength levels. Figure 11 visualises the per-replicate distributions of FDR (top), power (middle), and F1 score (bottom) across all nine cells of the factorial design. Corresponding tables for the Sparse and Polygenic architectures are provided in Table 8 and Table 9 in the Appendix.

**The joint determinant of power: per-marker effect size.** Reading across the nine cells of the factorial design reveals that no single parameter — neither $h_g^2$ nor $n_{\text{causal}}$ alone — determines detection performance. The true driver is their ratio, the per-marker effect size $h_g^2/n_{\text{causal}}$. At strong signal with sparse architecture ($h_g^2 = 0.35$, $n_{\text{causal}} = 10$; Table 8), each causal marker explains 3.5% of phenotypic variance, and lasso reaches a mean power of 0.420. At the same $h_g^2 = 0.35$ but with $n_{\text{causal}} = 100$ (Polygenic, Table 9), each marker explains only 0.35%, and lasso power collapses to 0.076 — a nine-fold reduction despite identical total heritability. Conversely, fixing $n_{\text{causal}} = 25$ and reducing $h_g^2$ from 0.35 to 0.10 produces an identical type of degradation: lasso power drops from 0.253 to 0.019 (Table 7). In both cases, the underlying quantity is the same: $h_g^2/n_{\text{causal}}$ falls below the threshold at which individual markers can be reliably distinguished from background LD noise across 6,523 tested positions.

This relationship is directly visualised in the accompanying interactive Shiny application, which exposes sliders for the proportion of variance explained (PVE, corresponding to $h_g^2$) and for $n_{\text{causal}}$. Moving either slider while holding the other fixed produces monotone shifts in the simulated power and FDR curves consistent with the factorial results reported here, providing an intuitive way to explore the parameter space beyond the nine fixed cells of Table 7. As a concrete illustration using the empirical heritability estimates from Table 3: setting the PVE slider to 0.80 (approximating lint percentage's estimated $\hat{h}^2 = 0.83$) produces visibly higher power for all methods compared with the fixed factorial cells, confirming that the simulation scenarios were deliberately conservative relative to the real trait heritabilities.

**Why the distributions look the way they do.** Several structural features of Figure 11 require explanation before interpreting the numbers. The scan-method FDR boxes (top panel) are extremely narrow and sit close to 1.0 in every cell. This near-zero variance is expected: when the selected set contains hundreds to thousands of markers, the FDR is determined almost entirely by the ratio of false positives to total selected markers, which is driven by the LD architecture of the genome rather than by the random seed. There is simply very little replicate-to-replicate variability in how many markers are in LD with a given set of causal loci. By contrast, lasso FDR boxes are wider and positioned lower, because lasso's small selected set (often fewer than 20 markers) means that whether any given replicate happens to recover one extra true positive shifts the FDR by a meaningful amount — this is numerical sampling variance, not instability. The stepwise boxes for FDR sit exactly at zero in the Moderate and Polygenic panels: when no markers are selected, FDR is undefined and is set to zero by convention; the absence of a box here means every replicate selected nothing, not that FDR was genuinely zero. Finally, the colour gradient from dark blue (strong signal) to light blue (weak signal) runs in the same direction across all methods and all metrics, confirming that $h_g^2$ shifts the entire distribution rather than just its mean.

**The central role of $h_g^2$ in determining detection performance.** The most striking pattern in Table 7 is the steep, monotonic decline in power as $h_g^2$ decreases from 0.35 (strong) to 0.10 (weak), with causal count held fixed at 25. For Lasso + PC + Season ($\lambda_{1\text{se}}$), mean power falls from 0.253 under strong signal to 0.109 under moderate and 0.019 under weak — a roughly 13-fold reduction. Both scan methods show the same monotonic pattern. This demonstrates that heritability is the primary driver of statistical detectability: when $h_g^2 = 0.10$, each causal marker explains on average only $0.10/25 = 0.4\%$ of phenotypic variance, which is too small to be reliably distinguished from the background noise of 6,523 correlated markers. In practical terms, reliably detecting loci of small effect requires either larger sample sizes, more stringent pre-selection of candidate regions, or relaxed thresholds accompanied by biological validation.

**FDR of scan methods is structurally bounded near 1.0 across all signal levels.** The top panel of Figure 11 shows that the FDR of the season-only and PCA-adjusted scans stays near 0.98–0.99 regardless of $h_g^2$. This is not because the scans become less accurate as signal weakens; rather, it reflects a structural property of BH-corrected marginal testing in high-LD data: the number of false positives scales with the total number of markers tested and remains large even as the number of true positives declines. In the weak-signal cell, the season-only scan selects approximately 283 markers on average (Table 7), but only $25 \times 0.185 \approx 5$ are expected to be truly causal. The scan FDR is therefore bounded away from zero by the LD architecture of the genotype matrix, independent of $h_g^2$.

**Lasso: the precision–power trade-off is most informative at moderate signal.** At strong signal ($h_g^2 = 0.35$), lasso selects on average 40.2 markers with power 0.253 and FDR 0.822 — far better than the scans' FDR of 0.983–0.988 — yielding F1 = 0.209 versus F1 = 0.032 and 0.024 for the PCA and season-only scans. At weak signal ($h_g^2 = 0.10$), lasso selects very few markers (mean 1.8) with high precision (0.469) but near-zero power (0.019), and its F1 of 0.036 is only marginally higher than the scans' 0.060–0.072. The moderate-signal cell ($h_g^2 = 0.20$) is the most informative comparison: lasso achieves F1 = 0.146 versus 0.082 and 0.037, confirming that lasso's precision advantage translates into a meaningful improvement in the balanced F1 metric when the signal is neither trivially strong nor too weak for any method to be useful.

**Stepwise selection fails in the Moderate architecture.** A notable result in Table 7 is that stepwise AIC selection consistently selects zero markers across all three signal levels (power = 0.000, N sel. = 0.000).

With 25 causal markers spread across 6,523 positions, the per-marker signal is too diffuse for most causal markers to rank in the top-50 by marginal $p$-value. The AIC criterion then finds no candidate that improves fit over the covariate-only null, and the procedure terminates without adding any markers. This is a genuine statistical outcome of the pre-filtering design, not an implementation error. As shown in Table 8, stepwise performs better in the Sparse architecture where 10 causal markers each explain a larger share of $h_g^2$ and are more likely to survive the top-50 filter.

**Architecture effects across all nine cells.** Comparing Table 8 and Table 9 with Table 7, power is consistently higher in the Sparse architecture and lower in the Polygenic architecture at the same $h_g^2$ level, because per-marker effect sizes shrink as causal markers are added. The middle and bottom panels of Figure 11 make this pattern visible: reading across the three architecture columns at the same fill colour, the median power and F1 both decline monotonically from Sparse to Moderate to Polygenic. This confirms that causal count and heritability are independent constraints on detection performance — precisely what the factorial design was constructed to disentangle.

# 5 Discussion

## 5.1 Summary and method recommendation

This report compared four approaches to QTL mapping in a high-dimensional cotton MAGIC population: a classical season-adjusted single-marker scan, a PC-adjusted variant of the same scan, a lasso multi-marker regression, and an AIC-guided stepwise forward selection. The empirical and simulation results lead to a clear and consistent conclusion about the trade-off between sensitivity and specificity.

The season-only and PC-adjusted scans are highly sensitive. They recover the majority of true causal markers across all three simulation scenarios and are especially effective when the genetic signal is strong and concentrated in a few loci. Their main limitation is specificity: they also select very large numbers of false positives, with FDR values consistently above 0.98. In practice, this means that a researcher who uses a BH-corrected scan to generate a candidate list will obtain a list dominated by non-causal markers that are merely correlated with the true causal loci through linkage disequilibrium. This output is informative for identifying broad chromosomal regions of interest, but it is not suitable for prioritising specific markers or genes for biological follow-up.

Lasso provides a better trade-off when the research objective is to obtain a short, interpretable set of marker candidates. Its FDR, while still non-trivial, is substantially lower than that of the scan methods, and its selected-set size is orders of magnitude smaller. The primary cost is reduced power: lasso misses more true causal markers, particularly under polygenic architectures where many small-effect loci must compete simultaneously under the $\ell_1$ penalty. However, the markers it does retain are much more likely to be genuinely associated.

The results also carry direct implications for cotton breeding. Lint percentage displayed the largest number of significant markers under all methods, consistent with its well-documented high narrow-sense heritability ($h^2 \approx 0.6$–$0.8$) in upland cotton populations. The strong negative phenotypic correlation between lint percentage and seed index ($r = -0.778$; see Figure 1) reflects a fundamental carbon source–sink trade-off in cotton boll development: assimilates directed toward lint fibre elongation are unavailable for seed filling, creating a yield–quality tension that breeders must navigate. Any genomic regions identified as jointly influencing both traits in opposite directions would represent particularly valuable targets for marker-assisted selection aimed at breaking this trade-off.

**Personal recommendation.** For a research workflow that aims to generate a small number of high-confidence candidate markers for biological validation, lasso under $\lambda_{1se}$ is the most appropriate primary analysis. If the goal is to characterise broad genomic regions for follow-up or for comparison with published QTL maps, the PC-adjusted scan provides a useful complement. Using both approaches together — as done here — is arguably the most informative strategy, because the Jaccard comparison between their selected sets identifies the subset of markers that are robust across fundamentally different modelling assumptions.

## 5.2 Comparison with existing literature on cotton QTL mapping

The patterns observed here are broadly consistent with findings from other multi-environment cotton QTL studies on MAGIC populations. Zitong's analysis of a closely related cotton MAGIC panel using interval mapping and mixed-model frameworks identified QTL regions for lint percentage and seed index that overlap with the chromosomal regions flagged by the scan methods in this report. The concentration of signal for lint percentage — by far the trait with the largest number of significant markers under all methods — is a recurring theme in the cotton genetics literature, where lint percentage is known to be strongly heritable and to map to a relatively consistent set of genomic regions across different genetic backgrounds and environments.

Two specific points of convergence with the findings of Li et al. (2018) are worth noting. First, the negative genetic correlation between lint percentage and seed index (overall $r = -0.778$; see Figure 1) is consistent with the antagonistic QTL effects reported at several loci in interval mapping studies, where alleles increasing lint percentage tend to reduce seed weight at the same genomic position, confirming that the phenotypic trade-off has a genetic basis rather than being purely environmental. Second, seed oil content, which showed the largest seasonal mean shift (Season 2 mean 18.38 vs Season 3 mean 21.68; Table 1), was also identified as environmentally sensitive in interval mapping analyses, suggesting that its QTL architecture includes loci with substantial genotype-by-environment interaction that single-environment regression models may not fully capture.

One notable point of divergence is methodological. Interval mapping, as applied by Zitong, identifies QTL peaks at chromosomal resolution by evaluating the likelihood of a QTL at every map position rather than at observed marker positions. This approach provides better localisation of QTL and implicitly accounts for the spacing between markers. The single-marker scan and lasso used here operate only at observed marker positions, so regions between markers are not evaluated. Direct comparison of selected markers to identified QTL intervals therefore requires checking whether lasso-selected markers fall within the confidence intervals reported by interval mapping, a step that is left for future work.

## 5.3 Limitations and future directions

Several limitations of this analysis deserve acknowledgement.

**Simulation design.** The current power analysis employs a $3 \times 3$ factorial design crossing causal marker count ($n_{\text{causal}} \in \{10, 25, 100\}$) with heritability ($h_g^2 \in \{0.35, 0.20, 0.10\}$), with season heritability fixed at $h_s^2 = 0.20$. This design is sufficient to show that both dimensions independently affect method performance, but the range of $h_g^2$ values could be extended, and the choice of top-50 pre-filtering for stepwise could be varied systematically to assess sensitivity. Increasing the number of simulation replicates from 30 to 100 or more would also reduce the Monte Carlo variance visible in the boxplot distributions.

**Marker correlation and linkage disequilibrium.** The single-marker scan does not account for the fact that markers within a QTL region are strongly correlated through linkage disequilibrium (LD). As a result, a single causal locus can produce a cluster of hundreds of nominally significant markers spanning several centimorgans. The Jaccard analysis shows that lasso partially resolves this by selecting one or a few representatives from each LD block, but the scan results in particular should be interpreted as indicating genomic regions rather than individual causal loci. The Manhattan plots in Figure 5 and Figure 6 make this LD clustering visually apparent as broad peaks rather than isolated spikes.

**Lack of biological follow-up.** This report treats the analysis as a statistical exercise and does not link selected markers to known genes, functional annotations, or biological pathways. The markers selected by lasso are worth annotating against the *Gossypium hirsutum* TM-1 reference genome to evaluate whether they fall near genes with known roles in fibre development (such as *GhMYB25-like*, *GhFSN1*, or *GhHOX3*) or in seed lipid biosynthesis. Such annotation would substantially increase the interpretive value of the statistical results and would connect the QTL signals to mechanistic hypotheses about fibre quality and seed composition.

**Model complexity.** All models in this report treat genotype effects as additive and do not consider dominance, epistasis, or genotype-by-environment interaction (GxE). In a MAGIC population where multiple parental haplotypes are segregating at each locus, modelling the full haplotype dosage rather than a collapsed biallelic score can provide additional power and resolution. Furthermore, the substantial seasonal shifts in seed oil content and seed index observed in Table 1 suggest that some QTL effects may differ across environments, which a GxE interaction model could formally test. These extensions are beyond the scope of the present report but represent important directions for more complete analyses of this dataset.

# 6 Appendix

## 6.1 Power analysis: Sparse and Polygenic architectures

The two tables below present the full factorial simulation results for the Sparse ($n_{\text{causal}} = 10$) and Polygenic ($n_{\text{causal}} = 100$) architectures, complementing Table 7 in the main text. All simulation settings (30 replicates, seed structure, four methods) are identical to those described in Section 3.8.

In the Sparse architecture (Table 8), stepwise AIC selection recovers a non-zero number of markers under the strong-signal condition ($h_g^2 = 0.35$), where 10 causal markers each explain a relatively large fraction of phenotypic variance and are therefore more likely to rank prominently in the top-50 pre-filter. Power and F1 are higher across all methods compared with the Moderate architecture at the same $h_g^2$ level, reflecting the larger per-marker effect size.

**Table 8.** Power-analysis summary — Sparse architecture ($n_{\text{causal}} = 10$). Each sub-panel shows results under a different signal strength ($h_g^2 \in 0.35, 0.20, 0.10$). Mean and SD over 30 simulation replicates.

| Method | Power | SD | Prec. | FDR | F1 | $N$ sel. |
|---|---|---|---|---|---|---|
| **Strong signal ($h_g^2 = 0.35$)** | | | | | | |
| Lasso + PC + Season ($lambda_{1se}$) | 0.420 | 0.154 | 0.309 | 0.691 | 0.356 | 17.567 |
| PCA-adjusted scan (BH) | 0.683 | 0.132 | 0.010 | 0.990 | 0.020 | 749.933 |
| Season-only scan (BH) | 0.677 | 0.128 | 0.007 | 0.993 | 0.014 | 1223.433 |
| Stepwise (AIC) | 0.240 | 0.110 | 0.269 | 0.731 | 0.254 | 9.533 |
| **Moderate signal ($h_g^2 = 0.20$)** | | | | | | |
| Lasso + PC + Season ($lambda_{1se}$) | 0.243 | 0.119 | 0.473 | 0.527 | 0.321 | 7.800 |
| PCA-adjusted scan (BH) | 0.490 | 0.163 | 0.072 | 0.928 | 0.125 | 257.267 |
| Season-only scan (BH) | 0.610 | 0.163 | 0.012 | 0.988 | 0.024 | 706.467 |
| Stepwise (AIC) | 0.210 | 0.118 | 0.230 | 0.770 | 0.220 | 9.800 |
| **Weak signal ($h_g^2 = 0.10$)** | | | | | | |
| Lasso + PC + Season ($lambda_{1se}$) | 0.057 | 0.073 | 0.561 | 0.263 | 0.103 | 1.133 |
| PCA-adjusted scan (BH) | 0.263 | 0.145 | 0.070 | 0.899 | 0.111 | 110.133 |
| Season-only scan (BH) | 0.370 | 0.166 | 0.024 | 0.976 | 0.045 | 276.700 |
| Stepwise (AIC) | 0.107 | 0.074 | 0.128 | 0.872 | 0.116 | 9.233 |

In the Polygenic architecture (Table 9), power is lower than in the Moderate and Sparse architectures at the same $h_g^2$ level, as each of the 100 causal markers contributes only a small fraction of the total genetic signal. Stepwise selection selects zero markers in all cells, for the same pre-filtering reason as in the Moderate architecture. The scan methods' FDR remains near 0.99 throughout, as large selected sets are dominated by linkage-disequilibrium neighbours of the true causal loci.

**Table 9.** Power-analysis summary — Polygenic architecture ($n_{\text{causal}} = 100$). Each sub-panel shows results under a different signal strength ($h_g^2 \in 0.35, 0.20, 0.10$). Mean and SD over 30 simulation replicates.

| Method | Power | SD | Prec. | FDR | F1 | $N$ sel. |
|---|---|---|---|---|---|---|
| **Strong signal** ($h_g^2 = 0.35$) | | | | | | |
| Lasso + PC + Season ($lambda_{1se}$) | 0.076 | 0.032 | 0.123 | 0.877 | 0.094 | 62.900 |
| PCA-adjusted scan (BH) | 0.299 | 0.083 | 0.033 | 0.967 | 0.060 | 946.733 |
| Season-only scan (BH) | 0.409 | 0.067 | 0.026 | 0.974 | 0.048 | 1672.467 |
| Stepwise (AIC) | 0.017 | 0.015 | 0.126 | 0.874 | 0.030 | 13.433 |
| **Moderate signal** ($h_g^2 = 0.20$) | | | | | | |
| Lasso + PC + Season ($lambda_{1se}$) | 0.017 | 0.019 | 0.129 | 0.697 | 0.031 | 15.500 |
| PCA-adjusted scan (BH) | 0.112 | 0.061 | 0.062 | 0.938 | 0.080 | 296.467 |
| Season-only scan (BH) | 0.234 | 0.079 | 0.028 | 0.972 | 0.050 | 912.933 |
| Stepwise (AIC) | 0.015 | 0.015 | 0.128 | 0.872 | 0.027 | 11.233 |
| **Weak signal** ($h_g^2 = 0.10$) | | | | | | |
| Lasso + PC + Season ($lambda_{1se}$) | 0.001 | 0.004 | 0.086 | 0.122 | 0.003 | 2.867 |
| PCA-adjusted scan (BH) | 0.020 | 0.027 | 0.062 | 0.719 | 0.031 | 47.267 |
| Season-only scan (BH) | 0.065 | 0.068 | 0.039 | 0.897 | 0.049 | 229.800 |
| Stepwise (AIC) | 0.007 | 0.009 | 0.059 | 0.941 | 0.013 | 12.533 |

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